

MOD-6

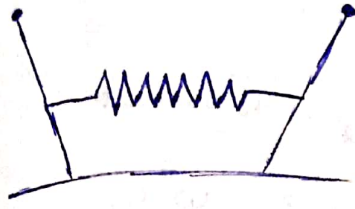
Two degree Freedom Systems

The system which requires two co-ordinates independent's to describe its motion ~~completely~~ is called a two degree freedom system. So two masses with two natural frequencies and two equations of motion.

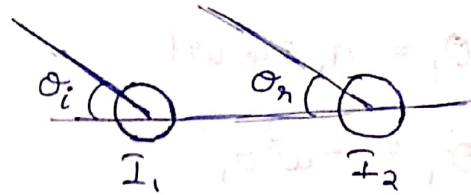
Principal modes of vibration

The system at its lowest or first natural frequency is called its first mode, at its next second highest is called second mode and so on. If the two masses vibrate at same frequency and in phase, it is called principal modes of vibration.

The system can be either rectilinear or angular as shown in fig.



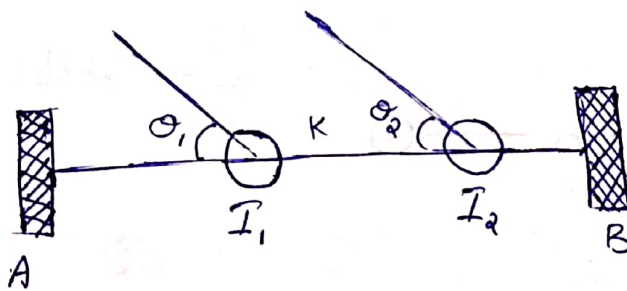
Rectilinear



Angular mode

(Torsional vibration)

Torsional vibrations



Consider shaft AB carrying two rotors having moment of inertia as I_1 & I_2 , θ_1 and θ_2 are angular displacement.

We have,

$$I_1 \frac{d^2 \theta_1}{dt^2} + K(\theta_1 - \theta_2) = 0$$

$$I_2 \frac{d^2 \theta_2}{dt^2} + K(\theta_2 - \theta_1) = 0$$

Let ,

$$\theta_1 = a_1 \sin \omega t$$

$$\ddot{\theta}_1 = -\omega^2 \theta_1$$

$$\theta_2 = a_2 \sin \omega t$$

$$\ddot{\theta}_2 = -\omega^2 \theta_2$$

$$-\omega^2 I_1 \theta_1 + K(\theta_1 - \theta_2) = 0$$

$$-\omega^2 I_2 \theta_2 + K(\theta_2 - \theta_1) = 0$$

so

$$-\omega^2 I_1 a_1 + K(a_1 - a_2) = 0 \quad \text{--- ①}$$

~~$$-\omega^2 I_2 a_2 + K(a_2 - a_1) = 0$$~~

$$-\omega^2 I_2 a_2 + K(a_2 - a_1) = 0 \quad \text{--- ②}$$

solving ,

$$\omega_1 = 0$$

$$\omega_2 = \sqrt{\frac{K(I_1 + I_2)}{I_1 \cdot I_2}}$$

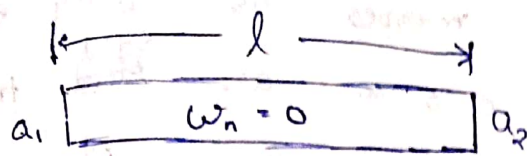
Put $\omega = 0$ in ① & ②

we get ,

$$a_1 = a_2$$

i.e.

$$\frac{a_1}{a_2} = 1$$



Also put $\omega = \sqrt{\frac{K(I_1 + I_2)}{I_1 \cdot I_2}}$ in ① & ②

we get

$$-\omega^2 I_1 a_1 = -K(a_1 - a_2) \quad \text{--- ③}$$

$$-\omega^2 I_2 a_2 = K(a_1 - a_2) \quad \text{--- ④}$$

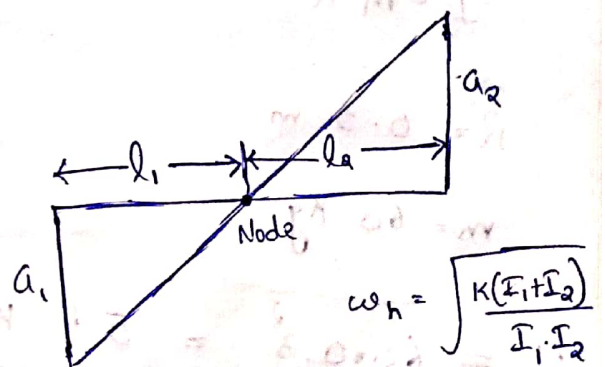
By dividing $\frac{\text{③}}{\text{④}}$,

we get

$$\frac{I_1 a_1}{I_2 a_2} = -1$$

$$\boxed{\frac{a_1}{a_2} = -\frac{I_2}{I_1}}$$

The section of the shaft where the angular displacement is zero is known as node.

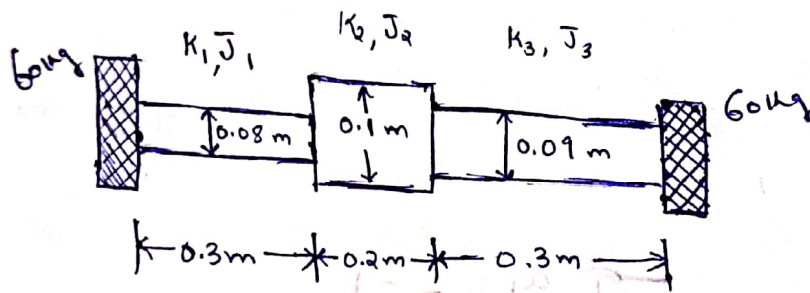


Q) Two bodies having equal masses as 60 kg each and radius of gyration 0.3 m are keyed to both ends of a shaft 0.8 m long. The shaft is 0.08 m in dia for 0.3 m length, ~~as also dia of~~ 0.10 m dia for 0.2 m length and 0.09 m dia for rest of the system.

Find the frequency of torsional vibration

($G = 9 \times 10^{11} \text{ N/m}^2$)

Ans)



$$I = m R^2$$

$$R = 0.3 \text{ m}$$

$$m = 60 \text{ kg}$$

$$\Rightarrow I = 60 \times 0.3^2 = \underline{\underline{5.4 \text{ kg m}^2}}$$

~~$$K_1 = \frac{G J_1}{L_1}$$~~

~~$$K_2 = \frac{G J_2}{L_2}$$~~

$$K_1 = G \cdot \frac{J_1}{l_1}$$

$$= 9 \times 10^{11} \times \frac{\frac{\pi}{32} \times (0.08)^4}{0.3}$$

$$= \underline{\underline{1.2057 \times 10^7 \text{ Nm/r}}}$$

$$K_2 = G \cdot \frac{J_2}{l_2}$$

$$= 9 \times 10^{11} \times \frac{\frac{\pi}{32} \times (0.1)^4}{0.2}$$

$$= \underline{\underline{4.415 \times 10^7 \text{ Nm/r}}}$$

$$K_3 = G \cdot \frac{J_3}{l_3}$$

$$= 9 \times 10^{11} \times \frac{\frac{\pi}{32} \times (0.09)^4}{0.3}$$

$$= \underline{\underline{1.93139 \times 10^7 \text{ Nm/r}}}$$

$$J = \frac{\pi}{32} \times d^4$$

$$\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3}$$

$$= \left[\frac{1}{1.2057} + \frac{1}{4.415} + \frac{1}{81.93139} \right] \times 10^{-7}$$

$$K = 6.356 \times 10^6 \text{ N m/r}$$

$$\omega = \sqrt{\frac{K(I_1 + I_2)}{I_1 I_2}}$$

$$I_1 = I_2$$

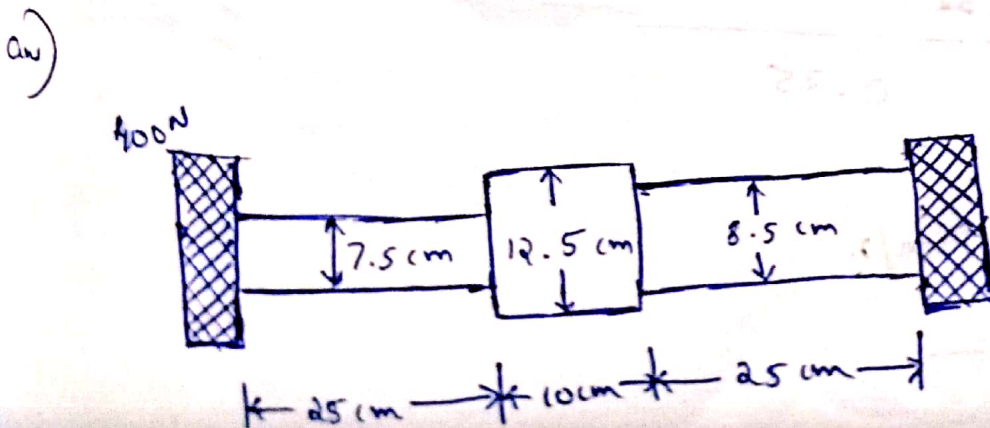
$$\Rightarrow \omega = \sqrt{\frac{2KI}{I^2}} \Rightarrow \sqrt{\frac{2K}{I}}$$

$$\Rightarrow \omega = 1.534 \times 10^3 \text{ r/s}$$

Torsionally Equivalent shaft

$$l = l_1 + l_2 \left(\frac{d_1}{d_2} \right)^4 + l_3 \left(\frac{d_1}{d_3} \right)^4 + l_4 \left(\frac{d_1}{d_4} \right)^4$$

Q) Two equal masses of weight 400 N each and radius of gyration 40 cm are keyed to the opposite ends of a shaft 60 cm long. The shaft is 7.5 cm dia for the first 25 cm of its length, 12.5 cm dia for the next 10 cm and 8.5 cm dia for the remaining of its length. Find the frequency of free torsional vibrations of the system and position of node. Assume $G = 0.84 \times 10^{11} \text{ N/m}^2$.



$$a) I = m R^2$$

$$m = \frac{W}{g}$$

$$W = 400 \text{ N}$$

$$g = 9.81 \text{ m/s}^2$$

$$I = \frac{400}{9.81} \times (40 \times 10^{-2})^2 \Rightarrow \frac{400}{9.81} \times (0.4)^2$$

$$I = \underline{\underline{6.524 \text{ kg m}^2}}$$

$$K_1 = G \cdot \frac{J_1}{l_1}$$

$$= 0.84 \times 10^{11} \times \frac{\pi}{32} \times (7.5 \times 10^{-2})^4$$

$$= \frac{0.84 \times 10^{11} \times \frac{\pi}{32} \times (7.5 \times 10^{-2})^4}{25 \times 10^{-2}} \text{ N m/rad}$$

$$= \frac{0.84 \times 10^{11} \times \frac{\pi}{32} \times (0.075)^4}{0.25}$$

$$= \underline{\underline{10.43 \times 10^5 \text{ N m/rad}}}$$

$$K_2 = G \cdot \frac{J_2}{l_2}$$

$$= 0.84 \times 10^{11} \times \frac{\frac{\pi}{32} \times (12.5 \times 10^{-2})^4}{10 \times 10^{-2}}$$

Nm/r

$$= 0.84 \times 10^{11} \times \frac{\frac{\pi}{32} \times (0.125)^4}{0.1}$$

$$= 201.2 \times 10^5 \text{ Nm/r}$$

$$K_3 = G \cdot \frac{J_3}{l_3}$$

$$= 0.84 \times 10^{11} \times \frac{\frac{\pi}{32} \times (8.5 \times 10^{-2})^4}{25 \times 10^{-2}}$$

$$= 0.84 \times 10^{11} \times \frac{\frac{\pi}{32} \times (0.085)^4}{0.25}$$

$$= 17.2 \times 10^5 \text{ Nm/r}$$

$$\frac{1}{K} = \frac{1}{K_1} + \frac{1}{K_2} + \frac{1}{K_3}$$

$$= \left[\frac{1}{10.43} + \frac{1}{201.2} + \frac{1}{17.2} \right] \times 10^{-5}$$

$$K = 6.29 \times 10^5 \text{ Nm/rad}$$

$$\omega = \sqrt{\frac{K(I_1 + I_2)}{I_1 \cdot I_2}}$$

$$I_1 = I_2$$

$$\Rightarrow \omega = \sqrt{\frac{2KI}{I^2}}$$

$$= \sqrt{\frac{2K}{I}}$$

$$\Rightarrow \sqrt{\frac{2 \times 6.29 \times 10^5}{6.524}}$$

$$\omega = 439 \text{ rad/s}$$

3) The position of the node is at

$$l_a = \frac{l_b I_b}{I_a}$$

Since, $I_a = I_b$

so, $\underline{l_a = l_b}$

so the node will lie in the middle of the equivalent shaft

4) Length of equivalent shaft is

$$l = l_1 + l_2 \left(\frac{d_1}{d_2} \right)^4 + l_3 \times \left(\frac{d_1}{d_3} \right)^4$$

$$= 0.25 + 0.10 \times \left(\frac{0.075}{0.125} \right)^4 + 0.25 \times \left(\frac{0.075}{0.085} \right)^4$$

$$= \underline{\underline{0.41 \text{ m}}}$$

Critical speed

When the natural frequency of a system coincides with external force frequency, it is called resonance. The speeds at which resonance occurs are known as critical speed.

At resonance, $\omega = \omega_n$

Damping factor, $\xi = \frac{C}{2m\omega_n} \Rightarrow \frac{C}{C_c}$

$\tan \phi = \alpha$, $\phi = -\frac{\pi}{2}$

Amplitude at resonance $= \frac{F}{C \cdot \omega_n}$

At critical speed, the amplitude of vibration of rotor excessively large and large amount of force is transmitted to the foundation. It may occur because of eccentric mounting of rotor, non-uniform distribution of rotor material, bending of shaft etc.

Q) A machine weighing 3.5 kg vibrates in a viscous medium. A harmonic exciting force of 40 N acts on the machine and produces a resonant amplitude of 18 mm with a period of 0.2 sec. Determine the damping coefficient.

aw) $m = 3.5 \text{ kg}$
 $F = 40 \text{ N}$

$$x_{ln} = 18 \text{ mm} \times 10^{-3}$$

$$T = 0.2 \text{ sec}$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\omega_n = 2\pi f$$

$$= 2\pi \times \frac{1}{T}$$

$$= 2\pi \times \frac{1}{0.2}$$

$$= \underline{\underline{31.41 \text{ rad/sec}}}$$

$$f = \frac{1}{T}$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$\Rightarrow 31.41 = \sqrt{\frac{k}{3.5}}$$

$$31.41^2 = \frac{k}{3.5}$$

$$K = 3454.36 \text{ N/m}$$

$$\Delta_n = \frac{F}{C \omega_n}$$

$$18 \times 10^{-3} = \frac{40}{C \times 31.41}$$

$$C = 70.74$$

Q) A shaft of 2.5 cm diameter, freely supported by bearing 75 cm apart, carries a single concentrated load of 196.2 N midway b/w the bearings. ~~Determine~~ Determine the first critical speed. Assume that shaft material has a density of $8 \times 10^3 \text{ kg/m}^3$ and E is $2.1 \times 10^{11} \text{ N/m}^2$.

$$\text{ans) } W_c = \sqrt{\frac{g}{\delta}}$$

~~$$\delta = \frac{W + \frac{17}{35} \times \rho \times l \times g}{48 E I}$$~~

$$\delta = \frac{\left(W + \frac{17}{35} \times \rho \times l \times g\right) \times l^3}{48 E I}$$

$$W = 196.2 \text{ N}, \quad l = 0.75 \text{ m}$$

$$\rho = 8 \times 10^3 \text{ kg/m}^3 \quad \left(\begin{array}{l} \text{convert } \rho \text{ into unit metre} \\ \text{so multiply with Area} \end{array} \right)$$

$$\Rightarrow \rho = 8 \times 10^3 \times \pi r^2$$

$$= 8 \times 10^3 \times \pi \times \left(\frac{d}{2}\right)^2$$

$$= 8 \times 10^3 \times \frac{\pi \times d^2}{4}$$

$$= 8 \times 10^3 \times \frac{\pi \times (2.5 \times 10^{-2})^2}{4}$$

$$= 8 \times 10^3 \times \frac{\pi \times (0.025)^2}{4}$$

$$\rho = 3.925 \text{ kg/m}$$

$$\boxed{I = \frac{\pi d^4}{64}}$$

$$I = \frac{\pi d^4}{64} \Rightarrow \frac{\pi \times (0.025)^4}{64}$$

$$I = \underline{\underline{1.916 \times 10^{-8} \text{ m}^4}}$$

So,

$$\delta = \frac{\left(W + \frac{17}{35} \times P \times \log \right) l^3}{48.6 \times I}$$

$$= \frac{\left(196.2 + \frac{17}{35} \times 3.925 \times 0.75 \times 9.81 \right) \times 0.75^3}{48 \times 2.1 \times 10^{11} \times 1.916 \times 10^{-8}}$$

$$= \underline{\underline{4.592 \times 10^{-4} \text{ m}}}$$

$$\omega_c = \sqrt{\frac{g}{\delta}}$$

$$= \sqrt{\frac{9.81}{4.592 \times 10^{-4}}}$$

$$= \underline{\underline{146.16 \text{ rad/s}}}$$

$$f_c = \frac{\omega_c}{2\pi} = \underline{\underline{23.27 \text{ Hz}}}$$

Q) A rotor of mass 12 kg is mounted in the middle of 25 mm diameter shaft supported b/w the bearings placed at 900 mm from each other. The rotor is having 0.02 mm ~~ecc~~ eccentricity if the system rotates at 3000 rpm. Determine the amplitude of steady state vibrations and the dynamic force ~~to~~ on the bearings.

$$\text{Take } E = 2 \times 10^5 \text{ N/mm}^2$$

Ans) $m = 12 \text{ kg}$, $d = 25 \text{ mm} \rightarrow 0.025 \text{ m}$, $E = 2.1 \times 10^5 \times 10^6 \text{ N/m}^2$
 $l = 900 \times 10^{-3} = 0.9 \text{ m}$, $e = 0.02 \times 10^{-3} = \text{m}$

$$K = \frac{48 EI}{l^3}$$

$$K = \frac{48 \times 2 \times 10^{5.6} \times \frac{\pi}{64} \times (0.25)^4}{(0.9)^3} \quad \left(I = \frac{\pi}{64} d^4 \right)$$

$$K = 252507.12 \text{ N/m}$$

Natural frequency,

$$\omega_n = \sqrt{\frac{K}{m}}$$

$$= \sqrt{\frac{252507.12}{12}}$$

$$= 145.05 \text{ rad/sec}$$

$$\omega = \frac{2\pi N}{60} \Rightarrow \frac{2\pi \times 3000}{60}$$

$$(N = 3000 \text{ rpm})$$

$$= 100\pi \text{ rad/sec}$$

$$\frac{x}{e} = \frac{1}{\left(\frac{\omega_n}{\omega}\right)^2 - 1}$$

$$\frac{x}{e} = \frac{1}{\left(\frac{145.05}{100\pi}\right)^2 - 1}$$

$$\frac{x}{e} = -1.270$$

$$x = -1.270 \times 0.02 \times 10^{-3}$$

$$x = -2.5 \times 10^{-5} \text{ m}$$

$$\text{Force} = K \times x$$

$$\Rightarrow 252507.12 \times 2.54 \times 10^{-5}$$

$$= 6.41 \text{ N}$$

$$\text{Load on each bearing} = \frac{6.41}{2} = 3.206 \text{ N}$$

Q) A rotor has a mass of 12 kg and is mounted midway on a 24 mm diameter horizontal shaft supported at ends by bearings. The bearings are 1 m apart. The shaft rotates at 2400 rpm. If the centre of mass of the rotor is 0.11 mm away from the geometric centre of the rotor due to certain manufacturing defects. Find the amplitude of steady state vibration.

Take $E = 200 \text{ GPa}$

ans) $m = 12 \text{ kg}$

$d = 24 \text{ mm} \Rightarrow 24 \times 10^{-3} = 0.024 \text{ m}$

$l = 1 \text{ m}$

$N = 2400 \text{ rpm}$

$e = 0.11 \text{ mm} \Rightarrow 0.11 \times 10^{-3} \text{ m}$

$E = 200 \text{ GPa} \Rightarrow 200 \times 10^9 \text{ N/m}^2$

$$k = \frac{48 EI}{l^3}$$

$$= \frac{48 \times 200 \times 10^9 \times \frac{\pi}{64} \times (0.024)^4}{(1)^3}$$

$$k = \underline{\underline{156345.785 \text{ N/m}}}$$

$$\omega_n = \sqrt{\frac{k}{m}}$$

$$= \sqrt{\frac{156345.75}{12}}$$

$$\omega_n = \underline{\underline{114.14 \text{ rad/sec}}}$$

$$\omega = \frac{2\pi N}{60}$$

$$\omega = \frac{2\pi \times 2400}{60} = \underline{\underline{251.32 \text{ rad/sec}}}$$

$$\frac{x}{e} = \frac{1}{\left(\frac{\omega_n}{\omega}\right)^2 - 1}$$

$$\frac{\gamma_c}{0.11 \times 10^{-3}} = \frac{1}{\left(\frac{114.14}{251.32} \right)^2 - 1}$$

$$\gamma_c = -1.38 \times 10^{-4}$$